CBC 2.0 • HACKATHON

Design Round

DocuSense AI — Extract. Validate. Verify. Correct.



TEAM NAME	TEAM ID	DOMAIN	COLLEGE
CACHE CODE	004	AI & ML	Sapthagiri NPS University

Problem Statement

PROBLEM STATEMENT (AS GIVEN)

Organizations receive invoices, application forms, certificates, and other documents in different layouts. Extracting information manually is repetitive and error-prone.

Objective: Build AI-powered system that extracts predefined information from uploaded documents and validates results.

Workflow:
- User uploads document or image.
- System identifies document type.
- Relevant fields extracted.
- Extracted values validated against expected formats/rules.
- Users review and correct uncertain fields.

Key Requirements:
- Support multiple document types.
- Extract configurable fields.
- Identify low-confidence or missing values.
- Validate dates, numbers, identifiers, and required fields.
- Allow users to correct extracted information.

Edge Cases:

PROBLEM ID

Problem Statement 3

DOMAIN

AI & ML

OUR FOCUS

Reliable extraction of configurable document fields, followed by validation and human review.

Understanding the Problem



Who is affected?

- • Operations, finance, HR, admissions, and document-verification teams.

• Staff handling invoices, certificates, and application forms.
• Organizations processing documents in different layouts.



What is going wrong?

- • Manual data entry is repetitive and error-prone.

• Different document layouts make rigid extraction unreliable.
• Missing, invalid, or conflicting values may be overlooked.



Why does it matter?

- • Errors can delay document processing and verification.

• Staff spend time on repetitive checking.
• Reviewers need traceable evidence to trust extracted values.

Our Proposed Solution



THE IDEA IN ONE LINE

DocuSense AI helps document-processing teams turn varied documents into structured, validated information through OCR, configurable field extraction, and human review.

How it works

1

Step 1 — Input

Users upload supported PDFs, scanned PDFs, or JPG/PNG images.

2

Step 2 — Process

The system reads text, identifies the likely document type, extracts configured fields, validates values, and flags missing, invalid, or conflicting information.

3

Step 3 — Outcome

Users inspect flagged fields, correct uncertain results, and save reviewed structured information.

Key Features

Feature 1 — Multi-Format Upload

Accept supported PDFs and images, and clearly reject unsupported or corrupted files.

Feature 2 — Document Classification

Identify likely document categories such as invoices, certificates, and application forms.

Feature 3 — Configurable Field Extraction

Use document-specific schemas to map relevant information into structured JSON.

Feature 4 — Validation Engine

Check required fields, dates, numbers, identifiers, and configurable rules.

Feature 5 — Confidence-Aware Review

Flag missing, uncertain, or conflicting values for human verification.

Feature 6 — Evidence-Linked Extraction

Proposed stretch feature: show source text, page references, and bounding-box highlights when OCR coordinates are available.

Architecture & Tech Stack



User layer

React + TypeScript + Tailwind CSS for web dashboard, upload, and review interface.



Application layer

Python + FastAPI for document upload, processing APIs, validation, review, and result retrieval.



Data & AI layer

PyMuPDF & OpenCV for parsing; PaddleOCR + optional LLM API; SQLite DB for prototype.

Tech stack

Design	Figma
Frontend	React / TypeScript
Backend	Python / FastAPI
Database	SQLite
AI / APIs	PaddleOCR / LLM API

Innovation & Differentiation

Aspect	Existing solutions	Our solution
Ease of use	Manual or rigid data-entry workflows.	One upload-and-review workflow.
Cost / access	Repetitive staff effort and tooling costs.	Accessible software stack for prototype; API limits apply.
Personalization	Layout-specific extraction rules required.	Configurable field schemas for different categories.
Scalability	Synchronous workflows become bottlenecks.	Asynchronous jobs & processing workers proposed for multiple uploads.

Our unique edge: DocuSense AI combines configurable field extraction, validation, conflict detection, and evidence-linked human review in one workflow.

Impact, Feasibility & Risks

Impact

Reduce repetitive data-entry work and help reviewers detect missing, invalid, and conflicting values. Actual time savings must be measured during testing.

Feasibility

A four-person team can target an end-to-end prototype using React, FastAPI, PaddleOCR, Python validation rules, and SQLite within the 18-hour implementation window.

Scalability

Use background jobs, job-status tracking, and worker processes. PostgreSQL, object storage, and a distributed queue are possible future upgrades.

Risks & mitigation

- Poor scans: preprocessing & quality warnings.
- Missing fields: explicit required-field flags.
- Conflicting values: retain alternatives & show source evidence.
- API failure: timeouts, safe retries & fallback.
- Sensitive documents: access controls & cautious API use.
- Large uploads: queued processing & performance testing.

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